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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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AUTONOMOUS CAR DETECTION USING COMPUTER VISION

Vision Pipeline, Camera Calibration & Depth Estimation

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INTRODUCTION



- An autonomous car, also known as a self-driving car, is a vehicle that can drive itself with little or no human intervention. It uses Artificial Intelligence (AI), Computer Vision, cameras, sensors, radar, LiDAR, and machine learning algorithms to understand its surroundings and make driving decisions safely.
- Computer vision enables the vehicle to recognize roads, detect pedestrians, identify traffic signs, avoid obstacles, and maintain safe navigation in real time.



COMPONENTS OF AN AUTONOMOUS CAR

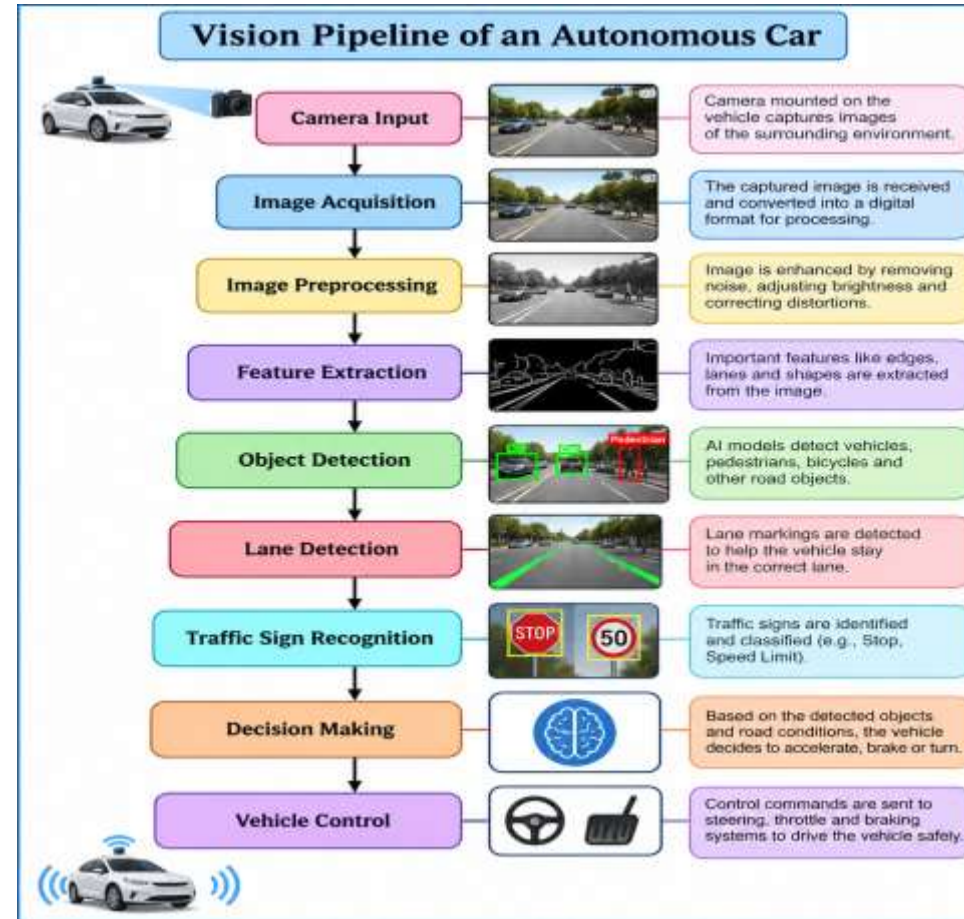


- An autonomous vehicle uses multiple sensors and technologies to perceive and understand its environment.

Component	Function
Camera	Captures road images for object and lane detection.
LiDAR	Measures accurate distances using laser beams.
Radar	Detects moving vehicles and measures their speed.
GPS	Provides the vehicle's current location.
Ultrasonic Sensors	Detect nearby obstacles during parking.



SYSTEM ARCHITECTURE





VISION PIPELINE



- **What is a Vision Pipeline?**
- A vision pipeline is a sequence of image-processing steps that enables an autonomous vehicle to understand the surrounding environment and make safe driving decisions.
- **Steps in the Vision Pipeline**
- **1. Image Acquisition**
- The camera continuously captures images of the road.
- **2. Image Preprocessing**
- The captured image is enhanced by reducing noise, adjusting brightness, and correcting distortions.
- **3. Feature Extraction**
- Important features such as road lanes, edges, traffic signs, and pedestrians are identified.



CAMERA CALIBRATION

What is Camera Calibration?

- Camera calibration is the process of determining the camera's internal and external parameters to remove image distortion and improve measurement accuracy.
- It helps the autonomous vehicle understand the real-world dimensions and positions of objects captured by the camera.

Why Camera Calibration is Important

- Removes lens distortion.
- Improves image quality.
- Provides accurate distance measurements.
- Enhances object detection.

DEPTH ESTIMATION

What is Depth Estimation?

- Depth estimation is the process of calculating the distance between the autonomous vehicle and surrounding objects.
- It allows the vehicle to determine how close or far objects are and helps avoid collisions.

Importance

- Collision avoidance.
- Safe braking.
- Distance measurement.
- Navigation.
- Path planning.



CONCLUSION

Autonomous vehicles rely heavily on computer vision to safely navigate roads. The **vision pipeline** processes images and identifies important objects, **camera calibration** improves image accuracy by correcting distortions, and **depth estimation** measures the distance to surrounding objects. Together, these technologies enable autonomous vehicles to detect roads, pedestrians, traffic signs, and obstacles, helping them make intelligent driving decisions and improving road safety.

THANK YOU!